Design Assignment 01

Figure 1 shows a layout diagram of a proposed factory building. Part of the building is assigned for the production area and other part for the office area. The height of the production area is 6m and it is 3m in the office area. The office area is to be fully air conditioned using 400 V, 16 kW, 0.85pf package type air condition. It is expected to install 6 numbers of 300W personals computers and one 500W photocopy machine in the office area too. (You could propose suitable arrangement for these equipment)

Following are the details of machines which are to be installed in the production area.

Machine	No. of Phases & Voltage	Rated Power output	p.f	Efficiency
MC1	3P, 400V	43.5kW	0.85	0.9
MC2	3P, 400V	18.5 kW	0.85	0.9
MC3	1P, 230V	5.5kW	0.85	0.87
MC4	1P, 230V	4.2 kW	0.85	0.85

There is a 10kW, 3P, 400V, 0.85p.f. water pump, around 90m away from this building, but have to be fed from the main panel of the building.

1. Propose suitable socket outlet arrangement, DB locations, and cable path for the machines.
2. Propose a suitable cable path for the machines and show them in a "Cable Layout Drawing".
3. Calculate the maximum demand for the factory and propose a suitable rating for the incomer.
4. Propose suitable cable size for each machine.
5. Design an approximate balanced three phase distribution system and present it as a "Single Line Diagram" (show the rating and type of protective devices, cable sizes, etc in your single line diagram).

Submission: A comprehensive report in PDF format incorporating all calculations, design drawings, and clearly stated assumptions adopted in the design

Note: This is an individual design assignment and must be submitted as a formal design report. Sharing of work with peers is strictly prohibited. Handwritten sections are not permitted. The use of AI tools for design generation or report preparation is strictly prohibited.

Deadline for submission: May 23, 2026 @ 11.59pm.

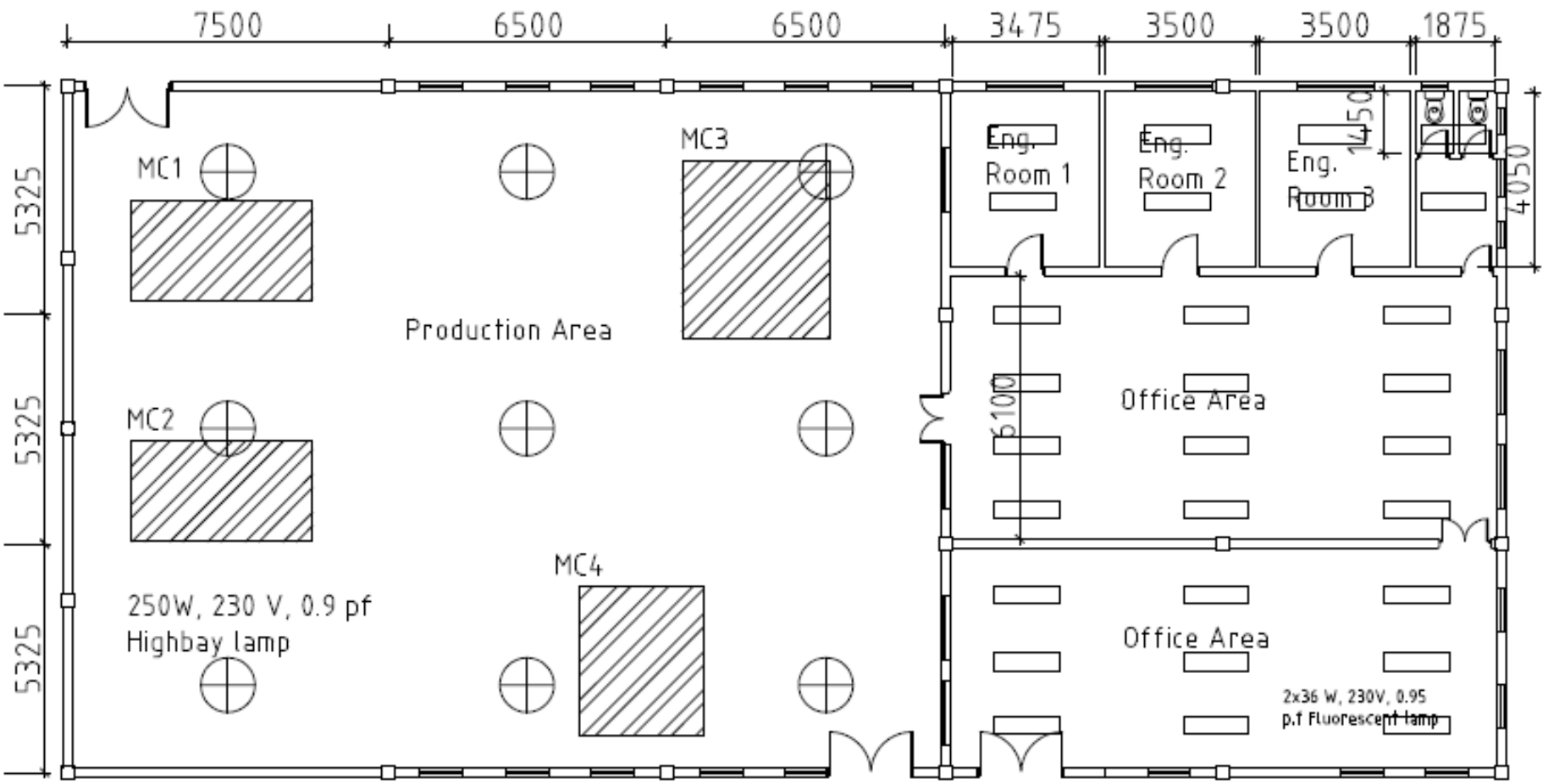


Figure 1 (All dimensions are in millimeters.)

Note: Original AutoCAD drawing is attached.